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The Development of Quality Management Information System under Quality 4.0 Era: A Review

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***Abstract*—Quality management (QM) practitioners pay close attention to the potential to unlock QM. In the context of the information age, information technology provides a new platform for QM. The relationship between QM and information systems (IS) are introduced in this study. The meaning of quality, quality management system (QMS) are explained respectively. The concept of information and how to collect, record, store, and report quality information are also introduced. Then, we analyze the relationship among information, digitalization, supply chain, performance, intelligence, manufacturing, sustainability, and Quality 4.0. We also analyzed how to grasp the advantages of quality management information system (QMIS) for enterprises and individuals. The importance of QMIS, how to achieve QMIS innovation and sustainable development are discussed at the end of this study.**

Keywords: QMS; IS; Digitization; Sustainable development

I. INTRODUCTION

QM is very important in the whole process of production; However, with the increase of production scale, the QMS has been impacted and challenged in terms of control of the production process, and we urgently need to explore better QM methods. At present, in Industry 4.0, many IS-related technologies are improving the production efficiency of products and reducing the quality cost of production from different perspectives [26]. Simultaneously, the shift to Industry 4.0 has given rise to many new conceptual vocabularies with the same affixes, such as Quality 4.0. In simple terms, Quality 4.0 can be understood as the digitalization of the QMS and the impact of information technology on quality technology, processes and individuals, and the approach of using modern technology to improve the seven tools of QM points to the direction of how to better use QM tools [20].Based on this, this study aims to understand how QM can become more efficient with the assistance of IS and how to make good use of their advantages brought by IS.

II. LITERATURE REVIEW

A. *Review of QMS*

For "QMS", we can split it into three words, namely "quality", "management" and "system". First of all, let's introduce "quality", quality is a relative concept, its meaning is the degree to which a set of intrinsic characteristics of an object satisfies the requirements, when comparing the intrinsic characteristics of things with the requirements, the degree of satisfaction obtained is called quality, and the higher this level of satisfaction, the better the quality. "Management," on the other hand, refers to the coordinated activities of the command and control organization, which includes the formulation of policies, objectives, and the process of achieving those objectives [24]. The command and control of the organizational coordination process is essentially the allocation of resources and activities by managers in the organization, as well as the actions of controlling the activities of the organization within a certain range and taking measures if necessary. Finally, there is the "system", which usually appears together with management here, constituting the "management system," which is the element of the interrelated and interacting process of the organization from the establishment of policies and goals to the realization of these goals. Returning to the concept of "QMS", it is the content in the quality management system, established by the organization, and necessary for quality objectives and QMS models [8][21]. In past data analysis, QM has been of great help to gradual product innovation and operational performance improvement of the organization.

B. The meaning of the information, the collection, recording, storage and reporting of quality information

In the QMS mentioned above, the word "information" is particularly important in QM. Information is meaningful data, and can be a variety of types or combinations, such as numbers, words, images, sounds, or standard samples. Whether it is the input, output, monitoring, measurement of the process, or the calculation and evaluation of the process performance, information is an indispensable medium, and we need to collect, record, store, and report the information and ensure its adequacy, appropriateness, and validity [1]. First of all, regarding how to collect quality information, Mr. Ishikawa listed five types of information, which are information that helps to understand the actual situation, analysis information, process control information, regulatory information, and acceptance or rejection information [5]. How to record and store, take the collection of customer needs as an example: Recording and storage can be divided into different categories at first, then extended customer requests or analysis of raw information, and finally useful quality information reported to interested parties [29].

C. The maturity of information technology

In fact, before the maturity of information technology, most of the quality information can only be processed by handwriting, printing, manual summary calculation, etc., and there is no computer assistance, which shows that the efficiency of the work is very low, and some human errors will occur. However, nowadays, computers are highly popular all over the world, and people can use computers to collect and process information, and even use AI for data analysis and prediction [2], and quality management information systems will be naturally summarized into computerized information systems. Therefore, the quality management information system includes both the quality management system and the information system, and we need to explore the relationship between the quality management system and the information system in more depth.

III. LITERATURE ANALYSIS

A. A model of the QMS process

In the increasingly fierce market competition, enterprises need to stand firm; They must pay enough attention to the QM of their products and services. Before we can do that, we must first familiarize ourselves with the basic processes of the QMS.

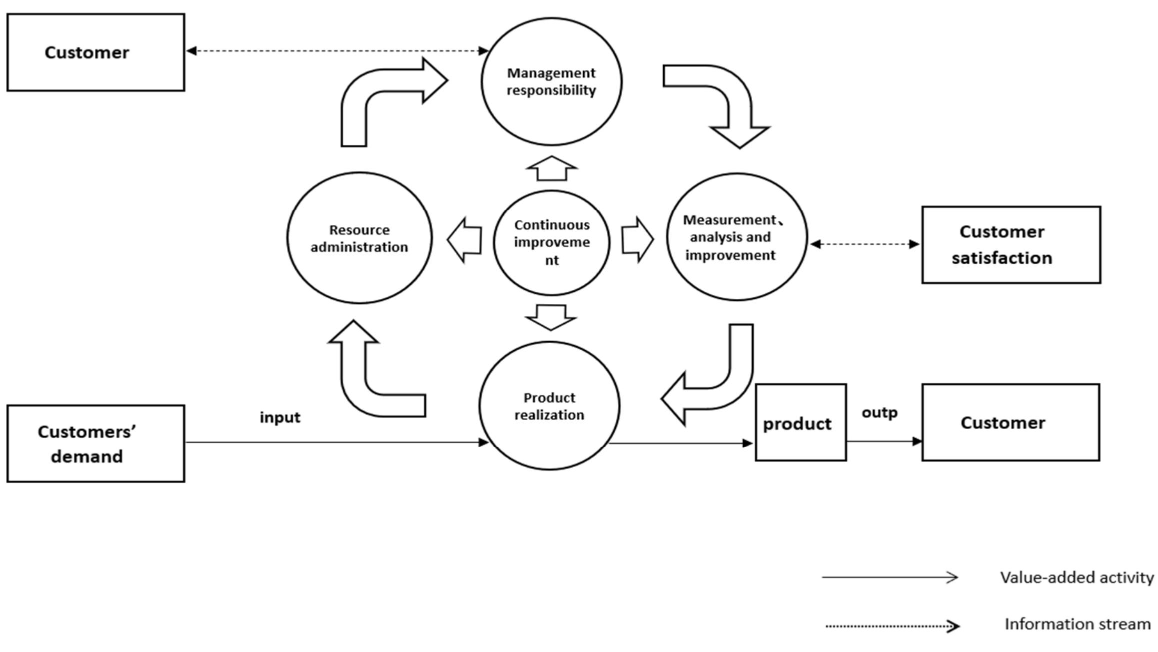

Figure 1. The basic process of the QMS

The basic concept of the QMS is the PDCA cycle of "plan-execute-check-action," as shown in Figure 1, able to continuously improve their own QMS according to customer feedback to meet the output of products.

B. Analysis of QM in IS

In the context of Industry 4.0, more and more academics are interested in the analysis of QM in IS. In the literature related to keywords such as Quality 4.0, Information, Digitalization, Supply Chain, Performance, Intelligence, Manufacturing, and Sustainability, we have compiled a number of articles related to these topics to explore how the QMS plays a role in IS and what the future trends will be. Therefore, among all the retrieved literature on the above topics, we preferentially selected literature covering a wide range of topics for statistical research, as shown in Table 1.

Table 1 Quality 4.0 Literature Review

Reference No.	Core Content
[1]	Project QM Cost Information System for Optimization of the Construction Industry
[10]	Leverage digitalization to strengthen supply chain QMS to improve performance
[11]	Technologies such as digitalization and Industry 4.0 play a positive role in industrial energy consumption and sustainable development
[12]	A literature review of ICTs, digitalization and environmental sustainability
[15]	The impact of QMS on intelligent transportation systems in the context of Industry 4.0 and digitalization
[16]	Supply chain QM is a key optimization for the manufacturing industry
[19]	Overview of sustainability and digitalization in QM
[23]	The link between QM and innovation and performance
[24]	The impact of supply chain and QM on the performance of non-large manufacturing firms in China
[25]	IS and Performance in Total Quality Management（TQM）

Table 2 Matrix analysis on Quality 4.0

Topic / Title	Supply Chain	Digitization	Performance	Quality 4.0	Manufacture	Information	Intelligent	Sustainability
[1]				√	√	√		
[10]	√	√		√				
[11]	√	√		√				
[12]		√				√		√
[15]		√		√			√	
[16]	√			√	√			
[19]		√		√				√
[23]			√	√				√
[24]	√		√	√				
[25]			√	√		√		

In the above-mentioned literature, we categorized and counted them literature from the eight dimensions of Quality 4.0, Information, Digitalization, Supply Chain, Performance, Intelligence, Manufacturing, and Sustainability, to make it easier to judge and analyze, and we present it in the form of a matrix, as shown in Table 2.

After the matrix analysis, we needed to know the number of mentions of topics in various dimensions to better understand which topics were most relevant to Quality 4.0. Therefore, we created a histogram, as shown in Figure 2, to count the number of mentions for each topic and arrange them sequentially.

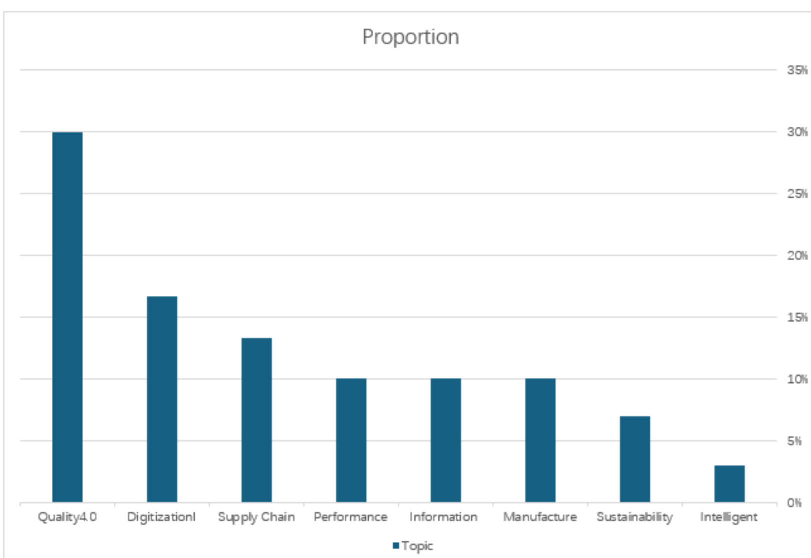

Figure 2. The number of mentions by topic

Table 3 Proportion and cumulative proportion of each theme

Topic	Occurrence number	Proportion	Cumulative Proportion
Quality4.0	9	30%	30%
Digitization	5	16.7%	46.7%
Supply Chain	4	13.3%	60%
Performance	3	10%	70%
Information	3	10%	80%
Intelligent	3	10%	90%
Manufacture	2	7%	97%
Sustainability	1	3%	100%

After intuitively counting the number of mentions of each dimension topic, we found that in addition to Quality 4.0, digitalization was the most mentioned topic. Therefore, we made a table of the proportion and cumulative proportion, as shown in Table 3, to understand the proportion of each topic more clearly.

Synthesizing the existing data for statistical analysis, we find that the topic of digitalization seems to be more related to Quality 4.0. How to view the changes brought by digitalization in the information system to the QMS, and how to prioritize the use of digitalization to bring economic benefits to QM, needs to be further discussed.

IV. DISCUSS

A. The development of QM in the context of Industry 4.0

In the above interactive analysis, it is not difficult to find that quality 4.0 is the most mentioned topic, and in fact, under the guidance of digitalization and other Internet technologies, the way of industrial production has undergone a huge transformation, so Industry 4.0 has begun to enter the public eye [7][9]. Due to the development of information technologies, for example artificial intelligence and big data and artificial intelligence, the traditional production model has changed, and the production system has become more efficient[30]. Similarly, traditional QM methods are increasingly difficult to meet the growing demands of mass production and manufacturing, and traditional QM methods have limitations, but Quality 4.0 is a spin-off of Industry 4.0 that innovates traditional QM methods [21][22]. It has taken our eyes to a wider field, whether it is production, education, public administration, transportation, communications, or even entertainment. As long as it involves the field of QM, information technology can be used to find more convenient QM methods, enhance competitiveness, and ensure sustainable development [6].

B. Digitalization and QM issues

In addition to Quality 4.0, the topic of digitalization also comes up very frequently, which shows that the relationship between digitalization and Quality 4.0 is very close. The truth is that digitalization is changing the world, and the impact of digital transformation will also permeate more industries, and it can open up new opportunities for businesses. At the same time, digitalization will also affect the global economy from the side, so wide and deep that it is difficult to distinguish between the various impacts. It is undeniable that digitalization has brought many new development opportunities for QM, but it has also brought more challenges. On the positive side, with the help of digitalization, companies or quality practitioners can more quickly carry out the process of detecting, diagnosing, and solving quality problems, reducing the time cost of previous manpower consumption, and improving internal efficiency and quality capabilities.in addition to this, digitalization has a significant impact on business models, operational processes, and customer experience[12].On the other hand, the benefits of digital transformation for SMEs are very limited, and only a small number of large companies can really benefit from digitalization, and SMEs may become victims of this, thus widening the distance between them and large enterprises. For most SMEs, the choice of digital technology is the primary issue, and whether it can ensure its continued operation and bring tangible benefits to the company in the future. companies and quality practitioners should also discuss how to tap the potential of digitalization at a deeper level and how to realize the practice of digital QM in a more comprehensive way[3][11].The changes brought about by contemporary digital technologies are affecting more and more fields, and organizations and individuals are changing the way information is collected and used [4].For example, throughout the supply chain, quality design, compliance, process control, and statistical analysis have prioritized the shift to digitalization or the use of advanced digital technologies to achieve production, R&D, and innovation[13].In conclusion, traditional QM techniques need to be broken and developed, and they need to receive more attention to stimulate endogenous motivation for QM. In the general direction of digital transformation, how enterprises and practitioners should grasp the tuyere is a matter of great concern.

C. QMIS with enterprises and practitioners

For enterprises, the company needs to set up an additional department responsible for information system management, because the processing and use of information can put you in a better position in the market competition, the management and use of IS will affect the company's decision-making to a certain extent, managers need to understand not only the capabilities of the organization, but also the constant focus on how information is being used in the organization. Research for SMBs in the market points out that prioritizing the introduction of digitalization into quality processes can help them transition to digital faster for a more volatile, complex, and innovative market. Similarly, through the development of digital leadership, it can effectively improve the innovation ability and innovation performance of enterprises, and create endogenous power for the long-term development of enterprises[13][22]. SMEs should be aware of these factors early on and use QMIS to thrive in emerging markets worldwide.

From the perspective of a QM practitioner, it is important to have the ability to choose, apply and use IS that are appropriate to the task and their responsibilities. In companies that have achieved world-class quality, most QM businesses have achieved a high degree of informatization and automation through the QMIS. These companies also guide the future development trend of the industry. Professionals working in QM must also understand the important aspects of ICT and Industry 4.0 and know how to effectively communicate information within the team[28]. At the management level, managers should also encourage organizational culture, lower the level of power culture, and emphasize individual and organizational quality initiatives to help a company achieve steady and long-term development in the future.

V CONCLUSIONS

Judging from the situation at home and abroad, this is a great era for human society entering the 21st century, which is always in the midst of change, and we can feel the major changes in the knowledge economy and information technology. With the introduction of a series of development strategies and policies on these new technologies, a large number of enterprises in China have given priority to enter the digital transformation, which has brought unprecedented opportunities for the R&D and application of management IS in China. At the same time, China's industry and information technology department pointed out that China is the country with the most complete industrial system in the world, and the added value of the manufacturing industry has ranked first in the world for more than twenty consecutive years since 2010, and the status of a major manufacturing country has been further consolidated. Among the 500 major industrial products, more than 40% of the products rank first in the world. However, in fact, the overall quality level still lags behind the growth of scale, and the situation of "big but not strong, complete but not excellent" has not fundamentally changed. Accelerating the high-quality development of the manufacturing industry, creating new competitive advantages for manufacturing brands, and promoting the leap of China's manufacturing industry to the global middle and high-end are still important strategic tasks in China's economic development at present and in the future.

In any sense, the development and progress in information technology are the engine of global economic and social change. Coupled with the process of globalization, fierce market competition has forced companies to improve their competitiveness to maintain their survival in a complex and volatile environment. The key to establishing the effective role of QM lies in how it is integrated with the continuous development of information technology[14].For example, modern management methods can reduce the loss of operations that do not add value, increase transparency, and create conditions for value-creating change. Or it is to introduce digital type innovation in the process, which allows information to be stored and shared, promotes information circulation, accelerates information search, protects its stability and continuity, helps grassroots employees work remotely, improves internal communication efficiency, expands new external communication channels, increases sales, and attracts new customers[16][17].In fact, under the influence of information technology, public policy has also entered a new era of digital leadership, which provides a lot of information about the behavior of different groups in society for the revision, promulgation and implementation of policies [27].

At the beginning of the 19th century, companies around the world tried to maximize their revenues, leading to a precipitous decline in huge reserves of natural resources, as well as the destruction of key resources such as the atmosphere, freshwater, and soil [18].As science and technology progressed, there was a greater understanding of resource dwindling and environmental biases, and by the 20th century, countries had placed greater emphasis on the theme of innovation for sustainable development. For example, digital and flexible supply chains play an active role in the positive impact on sustainability, and their role as intermediaries can flexibly adapt to market uncertainty and achieve desired results. It is undeniable that the sustainable development of all mankind contributes to the establishment of a more equitable, united, free, and tolerant society. The key to sustainable development lies in innovation-driven, using technological innovation in IS to promote the further development of QM, and ultimately bringing stable and sustainable output, this is a long-term issue worth discussing in today's world.

In general, the in-depth integration of QM and IS has a non-negligible role in the development of quality 4.0, and it is very likely to cause profound changes in the future industry.

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